

Scheduling (reliable triggers)

Audience: operators / engineers

Chinese version: SCHEDULING.zh-TW.md (./SCHEDULING.zh-TW.md)

GitHub's native schedule cron is often delayed or skipped. This project uses:

1. Primary: an external cron service calling the GitHub API (workflow_dispatch)
 2. Backup: sparse GitHub cron (Taiwan 10:30 / 15:30 / 18:30, weekdays)
 3. Manual: Actions UI or local scripts/trigger_update.sh
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1. Recommended external cron (cron-job.org, free)

1.1 Create a GitHub token

1. GitHub → Settings → Developer settings → Personal access tokens
2. Prefer a fine-grained token:
 - Repository access: only DownloadBankOfTaiwanExchangeRate
 - Permissions: Actions: Read and write
1. Copy the token (shown once)

Never commit the token. Store it only in the external cron service's secret fields.

1.2 One-command setup (when you have an API key)

1. Sign up at cron-job.org (<https://cron-job.org>) → copy API Key from Settings
2. Run locally (GitHub token uses gh auth token automatically):

```
export CRONJOB_API_KEY="your_cron_job_api_key"
python3 scripts/setup_cron_job.py
```

Creates two weekday jobs:

- 09:10–16:10 hourly
- 18:30 evening catch-up

Preview without creating:

```
python3 scripts/setup_cron_job.py --dry-run
```

1.3 Manual setup on cron-job.org (optional)

If you prefer the web UI:

Field	Value
Title	BOT FX Rates - Update
URL	https://api.github.com/repos/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions/workflows/update-rates.yml/dispatches

Schedule	see below
Request method	POST
Timezone	Asia/Taipei

Headers:

```
Accept: application/vnd.github+json
Authorization: Bearer <YOUR_GITHUB_TOKEN>
X-GitHub-API-Version: 2022-11-28
Content-Type: application/json
```

Body:

```
{"ref": "main"}
```

1.3 Suggested times (Taiwan)

09:10, 10:10, 11:10, 12:10, 13:10, 14:10, 15:10, 16:10, 18:30

Custom cron (if supported), timezone Asia/Taipei:

```
10 9-16 * * 1-5
30 18 * * 1-5
```

1.4 Success signals

- cron-job.org shows HTTP 204
- GitHub Actions shows a workflow_dispatch run for Update BOT FX Rates

2. Local Mac fallback (installable)

If your Mac is usually on, install a LaunchAgent (uses gh auth, no extra token):

```
./scripts/install_launchd.sh
```

Runs on weekdays at 09:10–16:10 and 18:30. Log: /tmp/bot-fx-trigger.log

3. Local / server crontab

```
# A) gh already logged in
./scripts/trigger_update.sh

# B) PAT
export GITHUB_TOKEN=ghp_xxx
./scripts/trigger_update.sh

# C) repository_dispatch
./scripts/trigger_update.sh --mode repository_dispatch
```

Example crontab:

```
10 9-16 ** 1-5 GITHUB_TOKEN=ghp_xxx /path/to/repo/scripts/trigger_update.sh >>/tmp/bot-fx.log 2>&1
30 18 ** 1-5 GITHUB_TOKEN=ghp_xxx /path/to/repo/scripts/trigger_update.sh >>/tmp/bot-fx.log 2>&1
```

4. GitHub backup schedule

UTC	Taiwan	Role
30 2 ** 1-5	10:30	backup
30 7 ** 1-5	15:30	backup
30 10 ** 1-5	18:30	backup

Treat the external cron as primary.

concurrency.group: update-bot-rates prevents overlapping runs from fighting each other.

5. Troubleshooting

Symptom	Action
401/403 from cron	Token missing Actions: write, or expired
404	Wrong workflow file name / repo path
204 but no Actions run	Workflow disabled, or wrong ref
Run OK but no commit	Rates unchanged (changed=false) — normal

6. Related files

- .github/workflows/update-rates.yml
- scripts/trigger_update.sh
- scripts/setup_cron_job.py (one-click cron-job.org setup)
- scripts/install_launchd.sh (Mac local fallback)
- TECHNICAL.en.md (./TECHNICAL.en.md)